

Products of the per-prime Tate curves of absolute geometry carry no correspondence calculus for the Weil explicit formula

Kunal Tyagi*

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Abstract

Connes and Consani’s June-2026 absolute geometry of $\text{Spec } \mathbf{Z}$ produces, for every prime p , a complex elliptic curve: the Tate curve $E_p = \mathbf{C}^\times / p^{\mathbf{Z}}$, a rectangular torus with period lattice $\mathbf{Z} \oplus \mathbf{Z} \cdot i \log p / (2\pi)$. A natural proposal is to assemble a correspondence calculus for the Weil explicit formula — the long-missing “square of $\text{Spec } \mathbf{Z}$ ” — from the (p, q) -graded family of classical abelian surfaces $E_p \times E_q$, importing Castelnuovo–Severi positivity fiber by fiber. We prove that the proposal is empty. The rigidity that does the first half of the work is not new and is not claimed here: for distinct primes, $\text{Hom}(E_p, E_q) \neq 0$ if and only if $\log p \cdot \log q \in 4\pi^2 \mathbf{Q}$ (Theorem 1), a transcendence coincidence impossible under the four-exponentials conjecture; unconditionally, each prime has at most one exceptional partner prime (Theorem 2), so among $E_2 \times E_3$, $E_2 \times E_5$, $E_3 \times E_5$ at most one surface carries a nonzero correspondence group (Theorem 3). **Theorems 1–3 are a specialization to primes of J. Winkelmann’s rigidity for the curves $\mathbf{C}^\times / \lambda^{\mathbf{Z}}$, from April 2002;** what is added here is the identification of the group $\text{Hom}(E_p, E_q)$, its rank and the isogeny degree, none of which appears in his paper. The new results are the consequences. Off this conjecturally empty matching, $\text{NS}(E_p \times E_q)$ has rank 2: there are no graph classes and no diagonal class, every divisor class acts as zero on H^1 , and the Castelnuovo–Severi inequality is vacuous (Theorem 4). On the diagonal, $\text{End}(E_p) = \mathbf{Z}$ unconditionally — the CM alternative $(\log p)^2 \in 4\pi^2 \mathbf{Q}$ is refuted by Gelfond–Schneider — and the complete intersection profile of the graphs Γ_m consists of universal integers independent of p (Theorem 5). No assignment of correspondence groups, composition data, and intersection pairings on the family realizes the required calculus (Theorem 6): three independent obstructions, each individually fatal. The family-level kill is unconditional; the paper carries a single open transcendence rider, whose negation is a published conjecture of D. Bertrand — the weak form of the four-exponentials conjecture — and which is consumed by none of the theorems.

Priority and attribution (please read before citing). Theorems 1, 2 and 3 below are anticipated. They are J. Winkelmann’s, from 15 April 2002 [Wi04]: studying $E_\lambda = \mathbf{C}^\times / \lambda^{\mathbf{Z}}$ for eigenvalues λ of elements of a discrete subgroup of $\text{SL}_2(\mathbf{C})$, he derives inside the proof of his Theorem 2 that isogeny of E_i and E_j forces $4\pi^2 / (\log \lambda_i \log \lambda_j) \in \mathbf{Q}$, divides two such relations to obtain $\log \lambda_i / \log \lambda_k \in \mathbf{Q}$ against multiplicative independence, runs the argument on three curves, and concludes that each curve in his family has at most one isogenous partner. Substituting $\lambda = p$ gives Theorems 1–3 here. His Lemma 7 is an equivalence, so the “iff” of Theorem 1 is his as well.

*hello.jay.tyagi@gmail.com. **Use of AI.** The mathematics in this paper — the derivations, the computations, the verification suite, and the text — was produced by Claude (Anthropic) working under the author’s direction inside a structured research program; the author set the program’s objectives, adjudicated its decisions, and is responsible for the content. Claude is a tool and is not an author. The complete program record, including every session log, review and correction, is public: <https://github.com/x67ai/riemann-rh-program>.

What is genuinely new in §§3–4 is only the prime-indexed explication: the group $\text{Hom}(E_p, E_q)$ identified, its rank, and the degree uv of the minimal isogeny — the words “Hom”, “rank” and “isogeny degree” do not occur in his paper, and he never takes λ prime. **The no-go itself — Theorems 4, 5 and 6, and the conclusion that products of the per-prime Tate curves host no correspondence calculus — is not anticipated and rests on no claim of priority over Theorems 1–3.** Two further corrections of record, both applied below: the per-prime elliptic-curve analogue in the Connes–Consani program dates from July 2015, not January 2025; and the rider (T1) is not, as an earlier draft said, unrecorded — it is the specialization to primes of the conjecture printed as (C4E faible) in [Dia97, p. 231], due to D. Bertrand.

0 Summary

Connes–Consani’s June-2026 absolute geometry of $\text{Spec } \mathbf{Z}$ [CC26] produces, for every prime p , a complex elliptic curve — the Tate curve $E_p = \mathbf{C}^\times/p^{\mathbf{Z}}$, a rectangular torus with period lattice $\Lambda_p = \mathbf{Z} \oplus \mathbf{Z} \cdot \tau_p$, $\tau_p = i \log p / (2\pi)$. For the first time, the absolute-geometry program yields per-prime genus-1 fibers carrying *classical* intersection theory. The seed proposal examined here was to assemble the long-missing correspondence calculus for the Weil explicit formula — the “square of $\text{Spec } \mathbf{Z}$ ” — from the (p, q) -graded family of classical abelian surfaces $E_p \times E_q$, importing the Castelnuovo–Severi positivity fiber-by-fiber instead of waiting for the unbuilt characteristic-one square.

This paper proves that the proposal is empty — not approximately, not “bandwidth data in geometric dress,” but empty. The results:

1. **Isogeny criterion (Theorem 1, iff):** for distinct primes $p \neq q$, $\text{Hom}(E_p, E_q) \neq 0$ **iff** $\log p \cdot \log q \in 4\pi^2 \mathbf{Q}$ — a transcendence coincidence expected never to occur (open unconditionally; impossible under the four-exponentials conjecture, §7). When nonzero, Hom has rank 1 and is computed exactly.
2. **Two-partner rigidity (Theorem 2, unconditional):** each prime has at most **one** “exceptional partner” prime with which it could share a nonzero Hom — pure unique factorization, no transcendence input. The exceptional pairs, if any exist at all, form a partial matching on the primes.
3. **The $\{2, 3, 5\}$ coherence kill (Theorem 3, unconditional):** among the three surfaces $E_2 \times E_3$, $E_2 \times E_5$, $E_3 \times E_5$, at most one carries a nonzero correspondence group; no two of them can simultaneously do so. Any composition-coherent correspondence family over all pairs of primes is unconditionally impossible.
4. **Néron–Severi collapse (Theorem 4):** $\text{NS}(E_p \times E_q) = \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2 \oplus \text{Hom}(E_p, E_q)$ [BL, Ch. 5], [RS14, Prop. 2.3]; reproved here. Off the exceptional matching, NS has rank 2 (the two fiber classes): there are **no graph classes, no diagonal class**, every divisor class acts as **zero** on H^1 , and the Castelnuovo–Severi/Hodge-index inequality — the positivity the seed was meant to be harvested “for free” — is **vacuous** (its primitive domain is the zero group).
5. **Diagonal residue (Theorem 5):** the surfaces $E_p \times E_p$ do carry a correspondence calculus — $\text{End}(E_p) = \mathbf{Z}$ **unconditionally**, the CM alternative $(\log p)^2 \in 4\pi^2 \mathbf{Q}$ being impossible by Gelfond–Schneider (§7) — but its complete intersection profile $(\Gamma_m \cdot \Gamma_n) = (m - n)^2$, $(\Gamma_m \cdot \Delta) = (m - 1)^2$ consists of **universal integers independent of p** . The explicit formula’s

transcendentally weighted prime terms $\Lambda(n)f(\log n)/\sqrt{n}$ cannot arise from it except by hand-inserting the weights into the test-family coefficients — re-entry into the known failure mode “no new data coordinate.”

6. **No-go (Theorem 6):** no assignment of correspondence groups, composition data, and intersection pairing on the family $\{E_p \times E_q\}$ realizes the required calculus for the Weil explicit formula. Three independent obstructions — no composition material (unconditional), no pairing target, prime-blind diagonal data — each individually fatal.

Of the two transcendence questions the construction raises, only one survives as a caveat: the diagonal coincidence $(\log p)^2 \in 4\pi^2\mathbf{Q}$ is **refuted unconditionally** (Gelfond–Schneider; §7, Proposition), and the off-diagonal coincidence $\log p \cdot \log q \in 4\pi^2\mathbf{Q}$ — (T1), an instance of the four-exponentials conjecture — is recorded as this paper’s single open rider (§7) with numerical evidence; **the family-level kill is unconditional and consumes neither.**

Scope (§8): this kills the *substrate family*, not the direction. The Connes–Consani characteristic-one square (route A) and the Deninger leg (route C) are untouched; this paper in fact sharpens why the square problem cannot be outflanked through the new complex fibers. Prior art (§9): nothing anticipates either the construction or the obstruction; seven adjacent traditions are cited and distinguished.

1 Background: the curves and the proposal

1.1 The missing square

The classical template for proving a Riemann hypothesis — Weil’s positivity argument on the square $C \times C$ of a curve over $\mathbf{F}_q$, with the Castelnuovo–Severi inequality supplied by Riemann–Roch and ampleness — needs a doubled object. Over $\mathbf{Q}$ that object (“the square of $\text{Spec } \mathbf{Z}$,” a surface-like host on which the Weil explicit formula becomes an intersection-number computation) exists in no branch of mathematics. Connes’ own summary of the stall, in print since 2015 and still accurate: “[*W*]hat is missing is an intersection theory and a Riemann–Roch theorem on the square of the arithmetic site” [CE15, §4.3.2]. The square does exist in characteristic one — Frobenius correspondences as congruences on the square of the arithmetic site [CC16] — but with no intersection theory and no Riemann–Roch, a decade on.

1.2 The per-prime Tate curves

The June-2026 paper [CC26] (“On the Absolute Geometry of $\text{Spec } \mathbf{Z}$ and the Fargues–Fontaine curve”) changes the available raw material. Evaluating the points of $(\text{Spec } \mathbf{Z})_{\mathbf{F}_1}$ over $\mathbf{C}$, the non-trivial points over a prime p form a torsor under the Weil group $W_\infty = \mathbf{C}^\times$; quotienting by the discrete Frobenius symmetries $p^{\mathbf{Z}} \subset W_\infty$ produces the **complex Tate curve**

$$E_p := M_p^\infty / p^{\mathbf{Z}} \cong \mathbf{C}^\times / p^{\mathbf{Z}},$$

“which canonically carries the structure of an elliptic curve over C ” [CC26, eq. (4) and §4.1.2]. Lifting through $w \mapsto \exp(2\pi iw)$ gives $E_p \cong \mathbf{C} / \Lambda_p$ with

$$\Lambda_p = \mathbf{Z} \oplus \mathbf{Z} \cdot \tau_p, \quad \tau_p = i \frac{\log p}{2\pi},$$

a rectangular torus defined over $\mathbf{R}$ (the paper’s own statement). Complex conjugation descends to a canonical real structure whose real locus is two copies of the adelic periodic orbit $C_p = \mathbf{R}_+^\times / p^{\mathbf{Z}}$. The

paper’s Geometric Decomposition theorem (stated as Theorem 4 in its introduction and Theorem 5 in §4.3 of the full text) gives a canonical global splitting

$$E_p \cong C_p \times \tilde{X}_\infty,$$

where C_p (length $\log p$) carries the entire dependence on p and the phase circle $\tilde{X}_\infty$ (length 2π , an unramified double cover of the real Fargues–Fontaine analogue $X_\infty \cong \mathbf{P}^1(\mathbf{R})$) is **totally independent of p** ; the modular parameter τ_p is exactly the ratio of the two invariant 1-forms’ periods. We use three facts and only these: the lattice Λ_p ; the reality/rectangularity; and the p -independence of the phase factor.

Lineage (first-appearance record). The per-prime *elliptic-analogue* idea appears in the Connes–Consani orbit on **21 July 2015**, in the Comptes Rendus note *The scaling site* [CC16b] and at length in *Geometry of the scaling site* [CC17]. The object there is $C_p = \mathbf{R}_+^\times/p^{\mathbf{Z}}$: a *real* circle of circumference $\log p$ — the periodic orbit of the scaling flow — carrying a quasi-tropical structure sheaf of convex piecewise-affine functions with slopes in $\mathbf{Z}[1/p]$, with Jacobian $J(C_p) \cong \mathbf{Z}/(p-1)\mathbf{Z}$ and a *real-valued* Riemann–Roch formula [CC16b, Lemma 6.3, Thm. 6.5, Thm. 6.7(ii)]. Its authors present it as an analogue and say so: “a quasi-tropical structure which turns this orbit into a variant $C_p = \mathbf{R}_+^\times/p^{\mathbf{Z}}$ of the classical Jacobi description $\mathbf{C}^*/q^{\mathbf{Z}}$ of an elliptic curve” [CC17, abstract].

What is new in June 2026 is not the per-prime analogue but the *complex* curve $E_p = \mathbf{C}^\times/p^{\mathbf{Z}}$, with a genuine complex structure, the lattice Λ_p , and classical intersection theory — with C_p reappearing inside it as the connected component of the real locus, $E_p \cong C_p \times \tilde{X}_\infty$ [CC26]. That is the object used here, and it is why the seed proposal became formulable at all. The intermediate paper [CC25] (January 2025) contributes a functor from finite abelian extensions of $\mathbf{Q}$ to finite covers of $X_{\mathbf{Q}}$, under which the monodromy of the periodic orbit of length $\log p$ corresponds to the Galois action of Frobenius at p ; it cites [CC17] for the elliptic-curve analogue rather than introducing it. None of these papers raises products of the E_p , correspondences between different primes, or any isogeny question — confirmed against the full texts (no isogeny/Hom/product-of-Tate-curves content).

1.3 The seed proposal being killed

The proposal under examination required, in the function-field template, a calculus with three components:

- **(R1) Doubled object + correspondence monoid.** Correspondence groups containing the diagonal Δ , the fiber classes ξ_1, ξ_2 , and one Frobenius-type class Ψ_p per prime; closed under composition and transpose; multiplicativity $\Psi_m \circ \Psi_n = \Psi_{mn}$ with Ψ_p primitive exactly at primes.
- **(R2) The pairing.** An intersection pairing with $(\Psi_f \cdot \Delta) = W(f)$ for test elements $\Psi_f = \int f(\lambda) \Psi_\lambda d^*\lambda$ — the Weil explicit formula as an intersection number, prime terms, archimedean term, and pole term all accounted.
- **(R3) Positivity.** A Hodge-index/Castelnuovo–Severi inequality on the host, derived from Riemann–Roch plus effectivity (a non-circular derivation, with no zeta input, is re-derived in a companion note; see the Provenance remark).

The seed proposed to realize (R1)–(R3) on the (p, q) -graded family of classical abelian surfaces $E_p \times E_q$, with the multiplicativity axiom realized as a descent/compatibility datum *across* the

family, and the archimedean and pole terms hoped to live on the $\tilde{X}_\infty$ factor. The proposal's own stated fear was that the family might only express *bilinear* data — $\Lambda(p^a)\Lambda(q^b)$ cross-terms, bandwidth-style data in geometric dress. Theorems 1–6 below show the situation is worse: the family expresses **no** prime-facing correspondence data at all.

For the structure of the target, recall the explicit formula in one standard normalization (Weil 1952 [We52]; the exact archimedean shape is immaterial here): for suitable test functions f , with $\hat{f}(s) = \int_{\mathbf{R}} f(x) e^{(s-1/2)x} dx$,

$$W(f) = \hat{f}(0) + \hat{f}(1) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (f(\log n) + f(-\log n)) - W_\infty(f),$$

W_∞ an explicit archimedean integral. Only three structural features are load-bearing for this paper:

- **(W-i)** the prime side is **linear** in $\Lambda(n)$, with transcendental weights $f(\pm \log n)/\sqrt{n}$ that vary with n and with f ;
- **(W-ii)** there is an archimedean term;
- **(W-iii)** there are pole terms.

2 Notation and the lattice presentation

Throughout, $p, q, \dots$ denote primes; set

$$y_p := \frac{\log p}{2\pi} > 0, \quad \tau_p := i y_p, \quad \Lambda_p := \mathbf{Z} \oplus \mathbf{Z} \cdot \tau_p, \quad E_p := \mathbf{C}/\Lambda_p.$$

Lemma 0 (lattice presentation; [CC26, §4.1.2]). $\exp_2: \mathbf{C} \rightarrow \mathbf{C}^\times$, $w \mapsto e^{2\pi i w}$, is a surjective homomorphism with kernel $\mathbf{Z}$, and

$$\exp_2(w + \tau_p) = e^{2\pi i w} \cdot e^{2\pi i \cdot (i y_p)} = e^{2\pi i w} \cdot e^{-\log p} = p^{-1} \cdot \exp_2(w).$$

Since $p^{\mathbf{Z}} = (p^{-1})^{\mathbf{Z}}$ as subgroups of $\mathbf{C}^\times$, $\exp_2$ induces an isomorphism $\mathbf{C}/\Lambda_p \cong \mathbf{C}^\times/p^{\mathbf{Z}}$. □

(Numerically confirmed to 55 digits — check A of the numerical suite; see the Provenance remark. The sign of τ_p is immaterial: $\mathbf{Z} \oplus \mathbf{Z}(-\tau_p) = \Lambda_p$.)

Standard facts used without further comment: every homomorphism of complex tori $\mathbf{C}/\Lambda \rightarrow \mathbf{C}/\Lambda'$ is induced by a unique $\lambda \in \mathbf{C}$ with $\lambda\Lambda \subseteq \Lambda'$ (lift to the universal cover); every holomorphic map between complex tori is a homomorphism followed by a translation (rigidity); $\text{Hom}(E, E')$ denotes the group of holomorphic homomorphisms.

3 Theorem 1: the isogeny criterion (iff)

Attribution (read before §§3–4). The rigidity proved in these two sections is Winkelmann's, from 15 April 2002 [Wi04], transported here to the per-prime Tate curves. It is not new and is not presented as new. Inside the proof of his Theorem 2 (v3, p. 16; v1, pp. 12–13) he proves exactly the three things proved below: the criterion, in the necessity direction — “Isogeny of E_i and E_j implies that there is a $\mathbf{Q}$ -linear relation between $4\pi^2$, $\log \lambda_i \log \lambda_j$, $2\pi i \log \lambda_i$ and $2\pi i \log \lambda_j$ (see lemma 7). Now $4\pi^2 \in \mathbf{R}$ and $\log \lambda_i \log \lambda_j \in \mathbf{R}$, while $2\pi i \log \lambda_i$ and $2\pi i \log \lambda_j$ are $\mathbf{Q}$ -linearly independent elements of $i\mathbf{R}$. Therefore a $\mathbf{Q}$ -linear relation can only exist [sic] if $4\pi^2/(\log \lambda_i \log \lambda_j) \in \mathbf{Q}$ ” —

with the converse one line from his Lemma 7, which is stated as an equivalence, so the “iff” is his too; the two-partner rigidity of Theorem 2, by dividing two such relations; and the three-curve run of Theorem 3, of which $\{2, 3, 5\}$ is the numerical instance. What is genuinely not in his paper, and is what these sections add, is the identification of the group $\text{Hom}(E_p, E_q)$, its rank, and the degree uv of the minimal isogeny: the words “Hom” and “rank” do not occur there, no isogeny degree is computed, and λ is never taken to be a prime. Proofs are given in full because the paper must be self-contained and because the prime-indexed forms differ in presentation from his.

Theorem 1. *Let $p \neq q$ be distinct primes. Then*

$$\text{Hom}(E_p, E_q) \neq 0 \iff y_p y_q \in \mathbf{Q} \iff \log p \cdot \log q \in 4\pi^2 \mathbf{Q}.$$

Moreover, when the condition holds and $y_p y_q = u/v$ in lowest terms ($u, v \in \mathbf{Z}_{>0}$), the full homomorphism group is

$$\text{Hom}(E_p, E_q) = \mathbf{Z} \cdot \lambda_0, \quad \lambda_0 = i v y_q,$$

of rank 1, and the isogeny induced by λ_0 has degree uv .

Proof. A nonzero homomorphism is multiplication by some $\lambda \in \mathbf{C}^\times$ with $\lambda \Lambda_p \subseteq \Lambda_q$. Two containment conditions:

1. $\lambda \cdot 1 \in \Lambda_q$: so $\lambda = a + b \cdot \tau_q = a + i b y_q$ for some $a, b \in \mathbf{Z}$.
2. $\lambda \cdot \tau_p \in \Lambda_q$: compute $\lambda \cdot \tau_p = (a + i b y_q) \cdot (i y_p) = -b y_p y_q + i a y_p$. Writing this as $c + i d y_q$ with $c, d \in \mathbf{Z}$:
 - real part: $-b y_p y_q = c \in \mathbf{Z}$;
 - imaginary part: $a y_p = d y_q$.

Case $a \neq 0$. Then $d \neq 0$ (as $y_p, y_q > 0$) and $y_p/y_q = d/a$, i.e. $\log p/\log q \in \mathbf{Q}$, i.e. $a \cdot \log p = d \cdot \log q$ with a, d nonzero integers, i.e. $p^a = q^d$ (after clearing signs) — impossible for distinct primes by unique factorization. So $a = 0$, hence $d = 0$.

Case $a = 0$. Then $\lambda = i b y_q$ with $b \neq 0$ (else $\lambda = 0$), and the surviving condition is $b y_p y_q \in \mathbf{Z}$, which forces $y_p y_q \in \mathbf{Q}$. This proves the forward direction, and identifies $\text{Hom} = \{i b y_q : b \in \mathbf{Z}, b y_p y_q \in \mathbf{Z}\}$. With $y_p y_q = u/v$ in lowest terms, $b u/v \in \mathbf{Z} \iff v \mid b$, so $\text{Hom} = \mathbf{Z} \cdot (i v y_q)$.

Converse. If $y_p y_q = u/v$, then $\lambda_0 = i v y_q$ satisfies $\lambda_0 \cdot 1 = v \cdot \tau_q \in \Lambda_q$ and $\lambda_0 \cdot \tau_p = (i v y_q)(i y_p) = -v y_p y_q = -u \in \mathbf{Z} \subseteq \Lambda_q$, so $\lambda_0 \Lambda_p \subseteq \Lambda_q$ and $\lambda_0 \neq 0$ — a genuine isogeny. (Numerically confirmed on a synthetic pair with $y y' = 3/7$ exactly — check B.)

Degree. $\text{covol}(\Lambda_p) = y_p$ and $\text{covol}(\lambda_0 \Lambda_p) = |\lambda_0|^2 \cdot y_p = v^2 y_q^2 \cdot y_p$, so $\deg = [\Lambda_q : \lambda_0 \Lambda_p] = v^2 y_q^2 y_p / y_q = v^2 \cdot (y_p y_q) = v^2 \cdot (u/v) = uv$.

Finally $y_p y_q \in \mathbf{Q} \iff \log p \log q = 4\pi^2 \cdot y_p y_q \in 4\pi^2 \mathbf{Q}$. □

Remark 1 (bookkeeping reconciliation). The criterion may be stated as $\log p \cdot \log q \in 4\pi^2 \mathbf{Q}$ or derived in the form $b \cdot \log p \cdot \log q = -4\pi^2 c$ and phrased as $\log p \cdot \log q \in \pi^2 \mathbf{Q}$. These are the **same condition**: $4\pi^2 \mathbf{Q} = \pi^2 \mathbf{Q}$ as subsets of $\mathbf{R}$, because $4 \in \mathbf{Q}^\times$ — a rational multiple of π^2 is a rational multiple of $4\pi^2$ and conversely. The normalization-free statement is $y_p \cdot y_q \in \mathbf{Q}$; we display the constant as $4\pi^2$ because $y_p y_q = \log p \cdot \log q / (4\pi^2)$ makes it the natural one.

Remark 2 (symmetry). The criterion is symmetric in p and q ; hence $\text{Hom}(E_p, E_q) \neq 0 \iff \text{Hom}(E_q, E_p) \neq 0$ (as the general theory also guarantees via the dual isogeny). “Exceptional partner” is therefore a symmetric relation.

4 Theorems 2 and 3: the unconditional family-level kill

Theorem 2 (two-partner rigidity). *For every prime p there is **at most one** prime $q \neq p$ with $\text{Hom}(E_p, E_q) \neq 0$. Consequently the set of “exceptional pairs” $\{p, q\}$ (pairs with nonzero Hom), if nonempty, forms a **partial matching** on the primes: no prime appears in two of them.*

Proof. Suppose $q \neq q'$ are two such primes. By Theorem 1, $y_p y_q \in \mathbf{Q}^\times$ and $y_p y_{q'} \in \mathbf{Q}^\times$. Dividing: $y_q/y_{q'} = (y_p y_q)/(y_p y_{q'}) \in \mathbf{Q}$, i.e. $\log q/\log q' \in \mathbf{Q}$, i.e. $q^m = q'^n$ for some positive integers m, n — impossible for distinct primes by unique factorization. $\square$

Nothing here consumes any transcendence input: the proof is unique factorization plus linear algebra over $\mathbf{Q}$ applied to the products $y_p y_q$, and it is agnostic about whether any exceptional pair exists.

Theorem 3 (the $\{2, 3, 5\}$ coherence kill). *Among the three surfaces $E_2 \times E_3$, $E_2 \times E_5$, $E_3 \times E_5$, at most one carries a nonzero correspondence group; in particular **no two of them** — **a fortiori not all three** — **can simultaneously carry nonzero correspondence groups**. (By Theorem 4 below, “correspondence group of $E_p \times E_q$ ” = $\text{Hom}(E_p, E_q)$.)*

Proof. Any two of the three pairs $\{2, 3\}$, $\{2, 5\}$, $\{3, 5\}$ share a prime, so two simultaneously nonzero groups would give some prime two exceptional partners, contradicting Theorem 2. Directly: if $y_2 y_3 = r_1 \in \mathbf{Q}^\times$ and $y_2 y_5 = r_2 \in \mathbf{Q}^\times$ then $y_3/y_5 = r_1/r_2 \in \mathbf{Q}$, i.e. $\log 3/\log 5 \in \mathbf{Q}$, i.e. $3^m = 5^n$ — impossible; the other two cases are identical. $\square$

Remark 3 (enlarging the family does not help). Define $E_n := \mathbf{C}^\times/n^{\mathbf{Z}}$ for any integer $n \geq 2$ (modulus $y_n = \log n/(2\pi)$) — the natural home for composite-index members Ψ_{pq} of the required monoid. The computation of Theorem 1 goes through verbatim, with one new phenomenon: the case $a \neq 0$ now requires $\log m/\log n \in \mathbf{Q}$, which for general integers means m and n are **multiplicatively dependent** ($m = k^a$, $n = k^b$), and then E_m, E_n are indeed isogenous (the subgroups $k^{a\mathbf{Z}}, k^{b\mathbf{Z}}$ of $\mathbf{C}^\times$ are commensurable). So the enlarged family admits exactly the isogenies of **one-generator towers** $\{E_{k^a}\}_a$ — and across towers, by the two-partner argument applied to multiplicatively independent moduli, at most the same conjecturally-empty exceptional structure as in Theorem 2, now a partial matching of *towers* (two partners n, n' of a fixed m are forced into a single tower, and a cross-tower isogeny would need the same unproven rational-product coincidence as Theorem 1). This is a structural echo of a parallel finding on the same axiom system: the printed polarized-Frobenius axioms are satisfiable precisely by one-generator (single-prime) monoids — the characteristic- p template is a one-prime template. The geometry says the same thing: per-prime towers exist; the cross-prime structure, which is where the arithmetic of ζ lives, is exactly what is absent.

Remark 4 (what a failure of the transcendence rider would and would not change). If some pair of primes actually satisfied $\log p \cdot \log q \in 4\pi^2 \mathbf{Q}$ — spectacular transcendence news — Theorem 1 would produce one isogeny of computable degree uv . It would not revive the seed: Theorems 2 and 3 cap the total supply of cross-prime maps at a partial matching, and the obstructions O2 and O3 of Theorem 6 are untouched.

5 Theorem 4: Néron–Severi collapse off the matching

We first fix the classical dictionary. For elliptic curves E, E' over $\mathbf{C}$, the group of *divisorial correspondences* from E to E' is $\text{NS}(E \times E')$ modulo the subgroup generated by the two fiber classes $\xi_1 = [\{0\} \times E']$, $\xi_2 = [E \times \{0\}]$ (Weil’s “trivial correspondences”):

$$\text{Corr}(E, E') := \text{NS}(E \times E') / \langle \xi_1, \xi_2 \rangle.$$

Theorem 4. *Let E, E' be elliptic curves over $\mathbf{C}$. Then there is a canonical isomorphism*

$$\mathrm{NS}(E \times E') \cong \mathbf{Z} \cdot \xi_1 \oplus \mathbf{Z} \cdot \xi_2 \oplus \mathrm{Hom}(E, E'),$$

hence $\mathrm{Corr}(E, E') \cong \mathrm{Hom}(E, E')$ ([BL, Ch. 5]; self-contained proof below). Consequently, for distinct primes $p \neq q$ with $\mathrm{Hom}(E_p, E_q) = 0$ (i.e., every pair outside the — conjecturally empty — exceptional matching of Theorems 1–2):

- (a) **Rank collapse.** $\mathrm{NS}(E_p \times E_q) = \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2$ has rank 2, with intersection form $\xi_1^2 = \xi_2^2 = 0$, $\xi_1 \cdot \xi_2 = 1$: every divisor class is numerically $a\xi_1 + b\xi_2$, and $\mathrm{Corr}(E_p, E_q) = 0$.
- (b) **H^1 -inertness.** Every divisor class on $E_p \times E_q$ acts as **zero** on H^1 (in both directions, $H^1(E_p) \rightarrow H^1(E_q)$ and the transpose). The correspondence calculus is not merely small; as a calculus of operators it is the zero calculus.
- (c) **No graphs, no diagonal.** The only holomorphic maps $E_p \rightarrow E_q$ are constants, so the only graph classes are fiber classes. There is no diagonal: not as a curve (a “diagonal” in a product of nonisomorphic curves means the graph of an isomorphism, and none exists), and not even as a numerical class — no $D \in \mathrm{NS}$ satisfies the diagonal’s numerical triple $(D \cdot \xi_1, D \cdot \xi_2, D^2) = (1, 1, 0)$, since $D \cdot \xi_1 = D \cdot \xi_2 = 1$ forces $D = \xi_1 + \xi_2$ and then $D^2 = 2 \neq 0$.
- (d) **Castelnuovo–Severi vacuous.** The Hodge-index/Castelnuovo–Severi inequality on a product of curves is the statement that the intersection form is negative semidefinite on the primitive part $\langle \xi_1, \xi_2 \rangle^\perp \subset \mathrm{NS}$. Here that primitive part is 0: if $D = a\xi_1 + b\xi_2$ satisfies $D \cdot \xi_1 = b = 0$ and $D \cdot \xi_2 = a = 0$ then $D = 0$. Equivalently: every class of bidegree (d_1, d_2) satisfies $D^2 = 2d_1d_2$ — Castelnuovo–Severi holds with equality everywhere, carrying zero information. The positivity the seed advertised as “free fiber-by-fiber” is free because there is nothing left for it to constrain.

Proof of the decomposition. All inputs are standard abelian-variety theory ([BL, Ch. 2] for the Poincaré bundle, seesaw, and rigidity; Ch. 5 for the endomorphism context; the statement itself is the classical correspondence dictionary). Write $\hat{E}' = \mathrm{Pic}^0(E')$ for the dual, P for the Poincaré bundle on $\hat{E}' \times E'$, normalized so that $P|_{\{\alpha\} \times E'} \cong \alpha$ for $\alpha \in \hat{E}'$ and $P|_{\hat{E}' \times \{0\}}$ is trivial.

Step 1 (the homomorphism φ_L). For $L \in \mathrm{Pic}(E \times E')$ put $L_x := L|_{\{x\} \times E'}$; $\deg L_x$ is constant in x . Normalize:

$$\tilde{L} := L \otimes \mathrm{pr}_1^*(L|_{E \times \{0\}})^{-1} \otimes \mathrm{pr}_2^*(L_0)^{-1},$$

so that $\tilde{L}_x = L_x \otimes L_0^{-1} \in \mathrm{Pic}^0(E')$ for all x , $\tilde{L}_0$ is trivial, and $\tilde{L}|_{E \times \{0\}}$ is trivial. By the universal property of $(\hat{E}', P)$ there is a unique morphism $\varphi_L: E \rightarrow \hat{E}'$ with $(\varphi_L \times \mathrm{id}_{E'})^*P \cong \tilde{L}$ (the a-priori ambiguity by pr_1 -pullbacks is killed by restricting to $E \times \{0\}$, where both sides are trivial). Since $\tilde{L}_0$ is trivial, $\varphi_L(0) = 0$; a pointed morphism of abelian varieties is a homomorphism (rigidity). So $\varphi_L \in \mathrm{Hom}(E, \hat{E}')$.

Step 2 (kernel; factorization through NS). $L \mapsto \varphi_L$ is additive: normalization and restriction are additive in L , and uniqueness plus the biadditivity of the Poincaré bundle ($((\mathrm{add}) \times \mathrm{id})^*P \cong \mathrm{pr}_{13}^*P \otimes \mathrm{pr}_{23}^*P$ on $\hat{E}' \times \hat{E}' \times E'$, [BL, Ch. 2]) give $\varphi_{L \otimes L''} = \varphi_L + \varphi_{L''}$. If $L \in \mathrm{Pic}^0(E \times E')$ then, since the dual of a product is the product of the duals, $L = \mathrm{pr}_1^*M \otimes \mathrm{pr}_2^*M'$ with $M \in \mathrm{Pic}^0(E)$, $M' \in \mathrm{Pic}^0(E')$; the normalization annihilates it, so $\varphi_L = 0$ and the map factors through $\mathrm{NS}(E \times E')$. Conversely if $\varphi_L = 0$ then $\tilde{L}_x$ is trivial for every x and $\tilde{L}|_{E \times \{0\}}$ is trivial, so by the seesaw principle $\tilde{L}$ is trivial; hence

$$L \cong \mathrm{pr}_1^*(L|_{E \times \{0\}}) \otimes \mathrm{pr}_2^*(L_0),$$

whose Néron–Severi class is $\deg(L|_{E \times \{0\}}) \cdot \xi_1 + \deg(L_0) \cdot \xi_2$ (the class of pr_1^* of a degree- d bundle is $d \cdot [\{\text{pt}\} \times E'] = d \cdot \xi_1$, and symmetrically). So the kernel of $\text{NS} \rightarrow \text{Hom}(E, \hat{E}')$ is exactly $\mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2$ (a free rank-2 group: the intersection numbers against ξ_2, ξ_1 read off the coefficients).

Step 3 (surjectivity and splitting). For $\psi \in \text{Hom}(E, \hat{E}')$ set $L^\psi := (\psi \times \text{id}_{E'})^* P$. Then $L_x^\psi = P|_{\{\psi(x)\} \times E'} = \psi(x) \in \text{Pic}^0(E')$, L_0^ψ is trivial, and $L^\psi|_{E \times \{0\}} = \psi^*(P|_{\hat{E}' \times \{0\}})$ is trivial — so L^ψ is already normalized and $\varphi_{L^\psi} = \psi$ by uniqueness. Additivity of $\psi \mapsto [L^\psi]$ is the Poincaré biadditivity again. This splits the sequence:

$$\text{NS}(E \times E') \cong \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2 \oplus \text{Hom}(E, \hat{E}').$$

Step 4 (eliminating the dual). For an elliptic curve, $x \mapsto [\mathcal{O}_{E'}([x] - [0])]$ is a canonical isomorphism $E' \cong \hat{E}'$ (the degree-one principal polarization; Abel–Jacobi), so $\text{Hom}(E, \hat{E}') \cong \text{Hom}(E, E')$. $\square$

Proof of (a)–(d). (a) is immediate from the decomposition with $\text{Hom} = 0$, plus $\xi_1^2 = \xi_2^2 = 0$ (disjoint parallel fibers) and $\xi_1 \cdot \xi_2 = 1$ (one transverse point).

(b) The action of a correspondence on cohomology, $\alpha \mapsto \text{pr}_{2*}([D] \cup \text{pr}_1^* \alpha)$, depends only on the class $[D] \in H^2(E \times E', \mathbf{Z})$. Künneth: $H^2 = (H^2 \otimes H^0) \oplus (H^1 \otimes H^1) \oplus (H^0 \otimes H^2)$. A class $\text{pr}_1^* \beta$ ($\beta \in H^2(E)$) acts on $\alpha \in H^1(E)$ by $\text{pr}_{2*}(\text{pr}_1^*(\beta \cup \alpha)) = 0$ since $\beta \cup \alpha \in H^3(E) = 0$; a class $\text{pr}_2^* \beta$ acts by $\beta \cup \text{pr}_{2*}(\text{pr}_1^* \alpha) = 0$ since fiber integration drops the degree of $\text{pr}_1^* \alpha \in H^1$ below zero. Now $\xi_1 = \text{pr}_1^*[\text{pt}]$ and $\xi_2 = \text{pr}_2^*[\text{pt}]$, so with $\text{NS} = \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2$ every *algebraic* class in H^2 has vanishing $H^1 \otimes H^1$ -component and acts as zero on H^1 . The transpose direction is identical using $\text{Hom}(E_q, E_p) = 0$ (Remark 2).

(c) Rigidity: every holomorphic map $E_p \rightarrow E_q$ is a homomorphism followed by a translation; $\text{Hom} = 0$ leaves the constants, whose graphs are horizontal fibers $E_p \times \{c\} \equiv \xi_2$. The numerical computation is displayed in the statement.

(d) Displayed in the statement; the classical Castelnuovo–Severi inequality itself (for products of curves with actual correspondences) is re-proved from Riemann–Roch + ampleness, with no zeta input, in a companion note (see the Provenance remark) — the engine is a real theorem whose domain of application is, on these surfaces, the zero group. $\square$

Remark 5 (what still exists on these surfaces). Rank-2 NS does not mean the surfaces are curve-poor: $\xi_1 + \xi_2$ is ample (Nakai: positive square, positive on every effective class), and general members of $|3(\xi_1 + \xi_2)|$ are irreducible curves dominating both factors. The point of (a)–(b) is that every such curve is *numerically* a fiber combination and acts as zero on H^1 : as a correspondence it is invisible. The collapse is a collapse of the correspondence calculus, not of the surface’s geometry — which is exactly what kills the seed, whose entire value proposition was the calculus.

6 Theorem 5: the diagonal residue is prime-blind — and the no-go

The one place the family genuinely has correspondences is the diagonal line $E_p \times E_p$.

Theorem 5 (diagonal residue). *Fix a prime p .*

1. $\text{End}(E_p) = \mathbf{Z}$, **unconditionally**. (The lattice computation below shows the only alternative would be $(\log p)^2 \in 4\pi^2 \mathbf{Q}$ — which would make τ_p imaginary quadratic and E_p a CM curve — and that coincidence is impossible: Gelfond–Schneider, §7, Proposition. No E_p has complex multiplication.)

2. By (1), $\text{NS}(E_p \times E_p) = \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2 \oplus \mathbf{Z} \cdot [\Delta]$ has rank 3 and $\text{Corr}(E_p, E_p) \cong \text{End}(E_p) = \mathbf{Z}$, generated by the diagonal. The available correspondences are (combinations of fiber classes and) the graphs Γ_m of multiplication by $m \in \mathbf{Z}$, with $\Gamma_1 = \Delta$, and their complete intersection profile is:

$$\Gamma_m \cdot \xi_1 = 1, \quad \Gamma_m \cdot \xi_2 = m^2, \quad (\Gamma_m \cdot \Gamma_n) = (m - n)^2 \quad (m \neq n), \quad (\Gamma_m \cdot \Delta) = (m - 1)^2$$

(the last for all m : at $m = 1$ this is $\Delta^2 = 0$, by adjunction with $K = 0$ and $e(E_p) = 0$).

3. **Every number in this profile is a universal integer, independent of p .** More precisely: under the real-affine identification $E_p \cong \mathbf{R}^2/\mathbf{Z}^2$ sending $(1, \tau_p)$ to the standard basis, the subtori Γ_m, ξ_1, ξ_2 correspond to the same subtori for every p , and intersection numbers of divisor classes are topological (cup products of integral classes). The map $p \mapsto$ (complete intersection profile of $E_p \times E_p$) is constant. The prime enters the surface only through its complex structure τ_p , and the complex structure enters the correspondence calculus only through Hom/End — which is the same ring $\mathbf{Z}$ for every p (part 1).

Proof. (1) $\lambda\Lambda_p \subseteq \Lambda_p$ with $\lambda = a + iby_p$ (from $\lambda \cdot 1 \in \Lambda_p$) requires $\lambda \cdot \tau_p = -by_p^2 + iay_p \in \Lambda_p$, i.e. $-by_p^2 \in \mathbf{Z}$ (and the τ_p -coefficient $a \in \mathbf{Z}$, automatic). If $b \neq 0$ then $y_p^2 \in \mathbf{Q}$, i.e. $(\log p)^2 \in 4\pi^2\mathbf{Q}$ — impossible by the §7 Proposition (Gelfond–Schneider; a $b \neq 0$ endomorphism would make $\tau_p^2 = -y_p^2 \in \mathbf{Q}$, i.e. τ_p quadratic imaginary and End a CM order, and no E_p admits this). So $b = 0$ and $\lambda = a \in \mathbf{Z}$: $\text{End}(E_p) = \mathbf{Z}$ with no exceptional case.

(2) Theorem 4 with $E = E' = E_p$ gives $\text{NS} = \mathbf{Z}\xi_1 \oplus \mathbf{Z}\xi_2 \oplus \text{End}(E_p)$, and the graph of id is Δ . Intersection numbers: Γ_m meets a vertical fiber $\{x\} \times E_p$ in the single transverse point (x, mx) , so $\Gamma_m \cdot \xi_1 = 1$. It meets a horizontal fiber $E_p \times \{c\}$ in the fiber $[m]^{-1}(c)$, which has m^2 points, all transverse ($[m]$ is unramified), so $\Gamma_m \cdot \xi_2 = m^2$; here $\#\ker([k]) = \#((\frac{1}{k}\Lambda_p)/\Lambda_p) = k^2$ (representatives $(i + j \cdot \tau_p)/k$, $0 \leq i, j < k$; check D of the numerical suite tabulates them for $k \leq 7$). For $m \neq n$, $\Gamma_m \cap \Gamma_n = \{(x, mx) : (m - n)x = 0\} = \ker([m - n])$, with $(m - n)^2$ points, each transverse (tangent directions $(1, m)$ and $(1, n)$ are independent), so $(\Gamma_m \cdot \Gamma_n) = (m - n)^2$; $n = 1$ gives $(\Gamma_m \cdot \Delta) = (m - 1)^2$ — which is also the Lefschetz number $1 - 2m + m^2$ of $[m]$ (trace on $H^0 - H^1 + H^2$), the integer Lefschetz data of the family.

(3) The subtori Γ_m are the images of $x \mapsto (x, mx)$, defined by the *integral* linear data m alone; under the identification by the basis $(1, \tau_p)$ they are the same integral subtori of $\mathbf{R}^4/\mathbf{Z}^4$ for every p . Cup products of integral classes on a fixed topological manifold are p -independent. $\square$

Theorem 6 (the no-go). *No assignment of correspondence groups, composition data, and intersection pairings on the family of classical surfaces $\{E_p \times E_q\}$ (p, q prime; optionally enlarged by composite moduli E_n per Remark 3) realizes the required calculus (R1)–(R3) for the Weil explicit formula. Specifically:*

- (O1) **The composition material does not exist (unconditional at family level).** (R1) needs one primitive Ψ_p per prime with composites $\Psi_p \circ \Psi_q$ supported across the family — nonzero elements of cross-prime correspondence groups for essentially all pairs. By Theorem 4, $\text{Corr}(E_p, E_q) \cong \text{Hom}(E_p, E_q)$; by Theorems 2–3 these groups vanish outside at most a partial matching of exceptional pairs, and already over $\{2, 3, 5\}$ at most one of the three groups is nonzero. The correspondence category of the family (objects E_p , morphisms Corr) is unconditionally too disconnected to carry a multiplicative monoid indexed by all primes: the multiplicativity axiom has no material to act on. (Whether even a single exceptional pair exists is the open transcendence question (T1) of §7 — unneeded for this conclusion.)

- (O2) **The pairing target does not parse off the diagonal.** (R2) pairs test correspondences against the diagonal. On every off-matching surface $E_p \times E_q$ there is no diagonal and no graph class — not even a numerical class with the diagonal’s profile (Theorem 4c) — and every class acts as zero on H^1 (Theorem 4b), so no trace-type functional is available; the Castelnuovo–Severi inequality (R3) is vacuous there (Theorem 4d). The positivity engine is a real theorem with an empty domain.
- (O3) **The diagonal residue cannot host the prime terms.** On the surfaces $E_p \times E_p$ the calculus exists but its complete intersection profile is the universal integer data of Theorem 5, identical for every p . A test element $\Psi_f = \sum_m c_m(p) \cdot \Gamma_m + a \cdot \xi_1 + b \cdot \xi_2$ has

$$(\Psi_f \cdot \Delta) = \sum_m c_m(p) \cdot (m-1)^2 + a + b,$$

a fixed integer-coefficient linear functional of the coefficients. By (W-i) the explicit formula’s prime side is linear in $\Lambda(n)$ with weights $f(\pm \log n)/\sqrt{n}$ — transcendental, n -dependent, f -dependent, and non-constant in p . Since the geometric data is constant in p , any dependence of $(\Psi_f \cdot \Delta)$ on p must be written into the coefficients $c_m(p)$ by hand — i.e., the designer inserts the answer key $\Lambda(p)f(\log p)/\sqrt{p}$ into the test family and the geometry contributes nothing. This is precisely the classical failure mode “Weil positivity in disguise” (no new data coordinate), the failure mode the proposal’s own terms promised to avoid: the prime contact was to be “the correspondence calculus itself or nothing.” It is nothing. The archimedean term (W-ii) fares no better: the candidate home named by the proposal — the phase factor $\tilde{X}_\infty$ — is p -independent by the source paper’s own decomposition theorem ([CC26, Thm. 5 and §4.3]: “totally independent of p ”), so it cannot source prime-dependent weights; and the pole terms (W-iii) have no candidate home at all.

Each obstruction is individually fatal; they are logically independent (O1 concerns the cross-prime category, O2 the off-diagonal surfaces, O3 the diagonal ones). $\square$

Remark (Rider: the escape routes forfeit the point). The p -flow and q -flow are conjugate through the real-analytic maps $\lambda e^{i\theta} \mapsto \lambda^{\log q / \log p} \cdot e^{i\theta}$ — but these are not holomorphic, and a smooth (or characteristic-one) correspondence theory on the products gives up exactly the classical $(1, 1)$ -class positivity (Castelnuovo–Severi via Riemann–Roch + ampleness) that was the seed’s entire selling point, landing back on the missing characteristic-one square with its absent intersection theory [CE15, §4.3.2]. There is no version of the family that keeps both the cross-prime maps and the classical positivity.

7 The transcendence status: (T2) refuted (Gelfond–Schneider), (T1) the single open rider

Two number-theoretic questions arise from Theorems 1 and 5 and are logged here. One of them turns out not to be open at all: classical transcendence theory refutes it, in this paper’s favor. The other is open — but it is a *named published conjecture*, not an unrecorded question, and an earlier draft of this paper said otherwise.

(T1) is published. Its negation is printed as a displayed, labeled conjecture — **(C4E faible)** — on p. 231 of Diaz [Dia97], attributed there to D. Bertrand (Madras Number Theory Symposium, January 1996; published as [Ber97, §5]): for positive real algebraic $\alpha_1, \alpha_2 \neq 1$, the numbers π^2 and

$(\log \alpha_1)(\log \alpha_2)$ are $\mathbf{Q}$ -linearly independent. Substituting $\alpha_1 = p$, $\alpha_2 = q$ gives exactly $\log p \cdot \log q \notin \pi^2 \mathbf{Q} = 4\pi^2 \mathbf{Q}$, i.e. $\neg(\text{T1})$; and the sentence immediately following it in Diaz is the same four-exponentials reduction performed below. His paper also carries unconditional partial results bearing on the question — his Théorème 1, the equivalence $(\text{C0}) \Leftrightarrow (\text{C6})$, and his Propositions 1 and 3(1) — which are not consumed here. Two riders. The English name “Bertrand’s weak four-exponentials conjecture” is not a term of art: Diaz’s label is (C4E faible), and Waldschmidt uses no such English phrase; call it the weak form of the four-exponentials conjecture, due to Bertrand. And the germ of the question is older still: Alaoglu and Erdős asked in 1944, in print, “If p and q are different primes, is it true that p^x and q^x are both rational only if x is an integer?” [AE44, p. 449], recording Siegel’s three-prime theorem on their p. 455 — the same four-versus-six-exponentials split rediscovered below.

Winkelmann’s own conjectural resolution is adjacent. His Conjecture 1 [Wi04, §3.2, p. 13] states that for algebraic α_1, α_2 with $|\alpha_i| > 1$ the quotients $\mathbf{C}^*/\alpha_i^{\mathbf{Z}}$ are isogenous iff α_1, α_2 are multiplicatively dependent; applied to distinct primes it gives $\text{Hom}(E_p, E_q) = 0$, hence $\neg(\text{T1})$ by the converse half of Theorem 1. He does not draw that consequence and never mentions primes, so the implication is recorded here as a one-line deduction from his conjecture and his Lemma 7, not as a result of his. He proves Conjecture 1 under Schanuel’s conjecture [Wi04, §3.2, Prop. 3, p. 14] — the same reduction performed below, seventeen years earlier.

- **(T1)** For distinct primes p, q : is $\log p \cdot \log q \in 4\pi^2 \mathbf{Q}$ possible? (Equivalently: can E_p and E_q be isogenous?) — **open unconditionally**; refuted under the four-exponentials conjecture (below).
- **(T2)** For a prime p : is $(\log p)^2 \in 4\pi^2 \mathbf{Q}$ possible? (Equivalently: can E_p have complex multiplication?) — **no; refuted unconditionally**:

Proposition (no CM anywhere in the family). *For every prime p , $(\log p)^2 \notin 4\pi^2 \mathbf{Q}$. Hence no E_p has complex multiplication, and $\text{End}(E_p) = \mathbf{Z}$ for every prime, unconditionally (Theorem 5(1)).*

Proof (Gelfond–Schneider, 1934). Suppose $(\log p)^2 = 4\pi^2 \cdot r$ with $r \in \mathbf{Q}$. Both sides are positive, so $r > 0$, and taking positive square roots, $\log p = 2\pi\sqrt{r}$, i.e. $p = e^{2\pi\sqrt{r}}$. With the branch $\log(-1) = i\pi$,

$$p = e^{2\pi\sqrt{r}} = e^{(-2i\sqrt{r}) \cdot (i\pi)} = (-1)^{-2i\sqrt{r}}.$$

The base -1 is algebraic, $\neq 0, 1$; the exponent $\beta = -2i\sqrt{r}$ is algebraic (a product of i and the real algebraic number $\sqrt{r}$), nonzero, and purely imaginary — hence irrational, whether or not $\sqrt{r}$ is itself rational. By the Gelfond–Schneider theorem, $(-1)^\beta$ is transcendental. But it equals the integer p — contradiction. $\square$

(Equivalently, in the linear-forms language: (T2) says $\log p = (-2i\sqrt{r}) \cdot (i\pi)$ — a $\overline{\mathbf{Q}}$ -linear relation between the two $\mathbf{Q}$ -linearly independent logarithms $\log p$ and $\log(-1) = i\pi$, one real and one imaginary; their ratio would be the algebraic irrational $-2i\sqrt{r}$, exactly what Gelfond–Schneider forbids. The point is that this *quadratic* relation among $\log p, \pi$ **factors** into linear-forms territory because $\sqrt{r}$ is algebraic. It is the classical family “ $e^{\pi\sqrt{d}}$ is transcendental” — the circle of Gelfond’s constant e^π and the Ramanujan constant — specialized to $d = 4r$.)

Status of (T1). Known unconditionally: $\log p / \log q \notin \mathbf{Q}$ (unique factorization — this is all Theorems 2, 3 consume), and the Proposition above on the diagonal. The genuinely quadratic *cross-prime* relation does not factor the same way — it involves two independent logarithms — and

no proven theorem currently excludes it. It is, however, an **instance of the four-exponentials conjecture** [4EC], not merely adjacent to it:

(4EC $\implies \neg(\text{T1})$). Suppose $\log p \cdot \log q = 4\pi^2 \cdot u/v$ ($u/v \in \mathbf{Q}$, necessarily positive). Take $x_1 = 2\pi i$, $x_2 = \log p$ — $\mathbf{Q}$ -linearly independent (compare real and imaginary parts) — and $y_1 = 1$, $y_2 = \log q/(2\pi i)$ — $\mathbf{Q}$ -linearly independent (y_2 is purely imaginary and nonzero, hence irrational). The four exponentials are $e^{x_1 y_1} = 1$, $e^{x_1 y_2} = q$, $e^{x_2 y_1} = p$, and $e^{x_2 y_2} = e^{\log p \cdot \log q/(2\pi i)} = e^{-2\pi i \cdot u/v}$, a root of unity. All four are algebraic — contradicting the four-exponentials conjecture. $\square$

The **proven** six-exponentials theorem does not obviously reach (T1), for an amusing reason: it needs a third row x_3 with both $e^{x_3 y_1}$, $e^{x_3 y_2}$ algebraic, and — taking $x_3 = \log \ell$ for a third prime, the natural candidate — the algebraicity of $e^{x_3 y_2}$ would itself require a *second* rational-product coincidence $\log \ell \cdot \log q \in 4\pi^2 \mathbf{Q}$, which this paper’s own Theorem 2 (two-partner rigidity) forbids alongside the first. The rigidity that makes the no-go unconditional is the same rigidity that keeps its one open rider out of the proven theorem’s reach.

Conditional resolution (Schanuel; retained as the stronger statement). Schanuel’s conjecture implies the full algebraic independence of $\log p, \log q, i\pi$ — hence $\neg(\text{T1})$, and it re-proves the Proposition. Take $x_1 = \log p$, $x_2 = \log q$, $x_3 = i\pi$. These are $\mathbf{Q}$ -linearly independent: in $a \cdot \log p + b \cdot \log q + c \cdot i\pi = 0$ the imaginary part forces $c = 0$, and then unique factorization forces $a = b = 0$. Schanuel then gives

$$\text{trdeg}_{\mathbf{Q}} \mathbf{Q}(\log p, \log q, i\pi, e^{x_1}, e^{x_2}, e^{x_3}) = \text{trdeg}_{\mathbf{Q}} \mathbf{Q}(\log p, \log q, i\pi, p, q, -1) \geq 3,$$

and since $p, q, -1$ are algebraic, $\log p, \log q, i\pi$ are algebraically independent over $\mathbf{Q}$. In particular $\log p \cdot \log q + r \cdot (i\pi)^2 \neq 0$ for every rational r — i.e. $\log p \cdot \log q \notin \pi^2 \mathbf{Q} = 4\pi^2 \mathbf{Q}$. $\square$

The status ladder for (T1). Open unconditionally $\subset$ settled by the four-exponentials conjecture (as an instance) $\subset$ settled by Schanuel (with full algebraic independence). Nothing in Theorems 2–6 consumes any rung of it: the family-level kill needs only unique factorization, which is exactly why it was engineered to be independent of (T1) — and (T2) is no longer a rider at all.

Numerical evidence (honesty check). Rows 1–3 of Table 1 are evidence on the open question (T1); rows 4–5 concern (T2), which the Proposition settles, and are retained not as evidence but as **detector controls confirming a theorem** — the instruments correctly find no rational value where Gelfond–Schneider says none exists. Working precision is 400 digits.

target	value (first 37 digits)	PSLQ (coeffs $\leq 10^{30}$)	CF terms to $q_N > 10^{100}$	max partial quotient
$\log 2 \cdot \log 3/(4\pi^2)$	0.0192890205998215679007346039996508659...	no relation	201	4591
$\log 2 \cdot \log 5/(4\pi^2)$	0.0282579044193212256166934388383576738...	no relation	201	1598
$\log 3 \cdot \log 5/(4\pi^2)$	0.0447877188535867805034209039422156347...	no relation	179	284
$(\log 2)^2/(4\pi^2)$	0.0121700170136801879558172286287261109...	no relation	160	226
$(\log 3)^2/(4\pi^2)$	0.0305723743263550882536179008172733061...	no relation	191	1521

Table 1: PSLQ and continued-fraction scans of the five targets at 400-digit working precision. The values are reproduced to their full recorded precision; the generating suite is named in the Provenance remark.

PSLQ finds no integer relation $m \cdot x + n = 0$ with coefficients up to 10^{30} for any target; the continued-fraction expansions, scanned until the convergent denominator exceeds 10^{100} , do not

terminate and show no anomalously large partial quotient — so **any rational value of any target has denominator exceeding** 10^{100} (if $x = a/b$ with $b \leq q_N$, its continued fraction terminates within the scanned, precision-guaranteed range). Detector control in the positive direction: the same PSLQ instance recovers $\log 8 = 3 \cdot \log 2$ instantly (check E). Status of this evidence: for (T1) it is evidence, not proof — and nothing in Theorems 2–6 consumes it; for (T2) the proof is the Proposition, and the numerics merely confirm the detectors work. In the exceptional-coincidence direction, Remark 4 records that even a true (T1) coincidence would create only a single isogeny inside the matching bound and would leave obstructions O2 and O3 intact.

8 Scope: what dies, what does not

What the no-go rules out.

- **The assembly itself.** The (p, q) -graded family $\{E_p \times E_q\}$ cannot carry (R1) — there is no composition material, unconditionally — nor (R2), there being no pairing target off the diagonal and only prime-blind data on it, nor a non-vacuous (R3).
- The planned “first empirical data on the seed” sandbox over $\{E_2 \times E_3, E_2 \times E_5, E_3 \times E_5\}$: ill-posed twice over — Theorem 3 shows no coherent correspondence assignment over the three surfaces exists to be measured (unconditionally, at least two of the three carry no correspondence class at all), and off the conjecturally empty exceptional matching the surfaces contain no graph classes and no diagonal to enumerate against (Theorem 4).
- The thesis that the family furnishes “a genuinely new substrate option the literature has never raised”: the option is genuinely new (§9) and genuinely empty.

What it leaves untouched.

- **The characteristic-one square of Connes and Consani.** Untouched — and complemented: the obstruction there is a *missing* intersection theory on an *existing* square [CC16][CE15, §4.3.2], whereas what is shown here is that the classical-complex escape from characteristic one — the one the June-2026 curves appear to open — is empty. The two results bracket the problem from opposite sides: the char-one square has correspondences (Frobenius congruences) but no positivity calculus; the complex surfaces have the full classical positivity calculus but no correspondences. What the explicit formula needs is *both at once*, and neither habitat supplies the pair.
- **Deninger’s foliated-dynamical route.** Untouched.
- **The Hodge index theorem over adelic curves.** Untouched, and if anything sharpened: the positivity engine is a real theorem, free of any Riemann-hypothesis input, but an engine is only as good as the substrate it runs on.
- **The curves E_p themselves.** Nothing here diminishes [CC26]’s construction; the no-go concerns only the pairwise *products* as a correspondence substrate — a use the paper itself never proposes.
- **Pair-indexed data as such.** The Kurokawa absolute-tensor-product line (§9, item 5) proves the *analytic* “square of Spec $\mathbf{Z}$ ” exists at the zeta-function level, with genuine Euler products over pairs of primes. Jointly with this paper, the failure is localized precisely: pair-of-primes data is realizable analytically; what is empty is the proposed *classical-geometric host* built from per-prime fibers.

Taxonomy. The result binds all per-prime-fiber substrate designs, with the following executable test for any future per-prime-fiber assembly proposal: (i) compute Hom between two named fibers by the one-page lattice criterion; (ii) exhibit the diagonal/graph classes the calculus needs on the actual NS group; (iii) check whether any intersection number can carry a $\log p$ weight without hand-insertion. The process lesson is that this gate — exactly the one-page computation of §3 — costs one page of lattice algebra and should run at proposal time.

9 Prior art: citations and distinctions

An earlier literature search found no source anticipating either the construction or the obstruction. **Its conclusion stands for the construction and is withdrawn for the obstruction.** The rigidity of Theorems 1–3 *is* anticipated, by Winkelmann in April 2002 (item 8 below, and the attribution note at the head of §3). The earlier claim that the obstruction “is elementary lattice arithmetic that could have been recorded about any family of tori with moduli $i \log p / (2\pi)$ and was not” is false: it was recorded, in 2002, and the reason the search missed it is that the search was phrased in prime-indexed language while Winkelmann’s is phrased in eigenvalue and geodesic-length language. What survives unanticipated is the construction — unsurprising, since the E_p are two months old with no citing literature — and the no-go proper, Theorems 4, 5 and 6. **Eight** adjacent traditions must be cited and distinguished:

1. **Connes–Consani–Marcolli, “The Weil Proof and the Geometry of the Adeles Class Space” [CCM07].** The correspondence-calculus-for-the-explicit-formula tradition — but on the square of the *one* global adèle class space, with correspondences built from graphs of the scaling action and analogs of degree/codegree (§7.1 there). Its §10.2 contains the germ of our objects: the tori $T_p \cong \mathbf{R}_+^\times / p^{\mathbf{Z}}$ as singular-fiber components in a vanishing-cycles analogy — one-dimensional *real* tori, no complex structure, no products, no intersection theory on products. **Distinction:** one global noncommutative space squared versus a (p, q) -family of classical surfaces; the no-go concerns only the latter. (The search also checked the distinct CCM paper “Fun with $\mathbf{F}_1$,” arXiv:0806.2401 — zero Tate/elliptic/product content.)
2. **Connes–Consani, “Geometry of the arithmetic site” [CC16] + the essay stall quote [CE15, §4.3.2].** The square *exists* in characteristic one, with Frobenius correspondences as congruences — and intersection theory/Riemann–Roch missing a decade on. **Distinction:** complementary result, as in §8: this note shows the classical-complex escape from that stall is empty, and the escape’s price (smooth/char-one correspondences) is the classical positivity itself.
3. **The lineage inside the Connes–Consani orbit [[CC16b, CC17] → [CC25] → [CC26]].** First appearance of per-prime elliptic-analogue language: **July 2015** — the periodic orbit $C_p = \mathbf{R}_+^\times / p^{\mathbf{Z}}$, a real circle of circumference $\log p$ with a characteristic-one structure sheaf, $J(C_p) \cong \mathbf{Z} / (p - 1)\mathbf{Z}$ and a real-valued Riemann–Roch [CC16b, CC17], described by its authors as “a variant . . . of the classical Jacobi description $\mathbf{C}^* / q^{\mathbf{Z}}$ of an elliptic curve”. [CC25] (January 2025) contributes the class-field-theory functor and cites [CC17] for the analogue rather than introducing it. The *complex* Tate curves are June 2026 [CC26], which never raises products, isogenies, or correspondences between primes (confirmed against the full text). **Distinction:** both the construction and the no-go of this note are additions to, not extractions from, that lineage — the lineage supplies per-prime fibers, never a cross-prime structure.

4. **Banaszak–Uetake [BU1, BU2, BU3].** Independent operator-theoretic axiomatization of a Weil-style intersection calculus (axioms INT1–INT3 on a Hilbert-space operator, modeled on Weil’s $C \times C$), whose *existence* is equivalent to RH (+ semi-simplicity). **Distinction:** they posit the calculus abstractly and prove existence $\iff$ RH; this paper kills the one concrete classical-surface instantiation ever proposed from per-prime geometric data. Opposite poles: abstract axioms with RH-equivalent existence versus concrete surfaces with provably empty calculus.
5. **The Kurokawa absolute tensor product line [Ku92, Ak09, Ta20].** Kurokawa’s 1992 proposal and the proven Euler products *over pairs of primes* for $\zeta \otimes \zeta$ and Dirichlet- L tensor squares (Koyama–Kurokawa; Akatsuka; Kurokawa–Wakayama; Tanaka) — “the square of $\text{Spec } \mathbf{Z}$ ” at the zeta-function level, zeros at sums $\rho_1 + \rho_2$. **Distinction and sharpening:** the pair-indexed *analytic* object exists and factorizes; the proposed *geometric* host from per-prime Tate curves is empty. Jointly the two localize the failure of the square at the geometric hosting step, not at pair-indexed data. (This line is also the published calibration for what a pairwise-product family *would* express if it expressed anything: pair-supported data with summed zeros — not $W(f)$.)
6. **Haran’s arithmetical-surface program [Ha91].** The closest non-Connes–Consani surrogate proposal: $\text{Spec } \mathbf{Z} \times_{\text{alg}} \text{Spec } \mathbf{Z}$ reduces to the diagonal (Haran: “the surface reduces to the diagonal!”), so Haran proposes *defining* intersection numbers of “Frobenius divisors” on the nonexistent surface by analytic means, aiming at a two-dimensional Riemann–Roch. **Distinction:** no per-prime elliptic fibers, no products of classical curves, no obstruction statement; Haran’s program postulates the calculus’s output, ours tests a concrete candidate host and finds it empty.
7. **Scholze A_{inf} /Fargues–Fontaine adjacency.** Nothing in that school produces cross-prime products of per-prime objects as a correspondence substrate (shtuka legs and X_S -families are all single- p), and no such no-go is published there. The one genuine contact runs through [CC26] itself: the phase factor $\tilde{X}_\infty$ as a *real analogue* of the Fargues–Fontaine curve — a per-prime-to-archimedean bridge, not a cross-prime one. **Distinction:** disjoint mechanism; cited for completeness of the search record.
8. **Winkelmann’s rigidity [Wi04].** Inside the proof of his Theorem 2 he derives, as the necessary condition for isogeny of $\mathbf{C}^*/\lambda_i^{\mathbf{Z}}$ and $\mathbf{C}^*/\lambda_j^{\mathbf{Z}}$, that $4\pi^2/(\log \lambda_i \log \lambda_j) \in \mathbf{Q}$; he then divides two such relations to force $\log \lambda_i / \log \lambda_k \in \mathbf{Q}$, runs the argument on three curves, and concludes that each curve in his family has at most one isogenous partner. That is §§3–4 of this paper, transported. **Distinction:** his context is the geometry of $\text{SL}_2(\mathbf{C})/\Gamma$ and hyperbolic geodesic length spectra; he never takes λ prime, never writes Hom , never computes a rank or an isogeny degree, and raises no Néron–Severi group, no correspondence calculus and no explicit formula. This note adds the prime-indexed explication and the no-go; it claims no priority over the rigidity.
9. **Also checked, and distinguished.** Connes–Consani’s Riemann–Roch strategy [CC19, §§6.4–6.6] carries their moduli of triangular elliptic curves under the equivalence generated by isogenies — a moduli-level construction, not a correspondence calculus on a product. Their “Knots, primes and the adèle class space” [CC24a, §3] does build a genuine two-prime object, $\Gamma \backslash (\mathbf{Q}_p \times \mathbf{Q}_q \times \mathbf{R})$ with $\Gamma = \{\pm p^m q^n\}$; **Distinction:** a knot/linking-number analogy, not a correspondence calculus. Morishita [Mor25] relates Deninger’s foliated dynamical systems to the Connes–Consani adelic spaces, and the surfaces here sit on neither. Milne’s survey [Mil16]

is the standard account of the Weil-proof tradition whose transport to the integers the seed proposal attempted.

Transcendence background (rider citations). The Gelfond–Schneider theorem (1934) for the **unconditional refutation of (T2)** — the diagonal coincidence factors into a $\overline{\mathbf{Q}}$ -linear relation between $\log p$ and $i\pi$, squarely inside the classical theorem (§7, Proposition); Baker’s theorem for the general linear-forms background; the four-exponentials conjecture [4EC] — posed by Schneider as Problem 1 of [Sch57, Ch. V, §4], formulated by Ramachandra [Ram68, II, §4] and stated as Conjecture 3.7 of Waldschmidt [Wal04] — of which **(T1) is an instance** (not merely an adjacent open problem), with the proven six-exponentials theorem (Lang [Lan66]) kept out of reach by Theorem 2 itself (§7); Schanuel’s conjecture for the stronger algebraic-independence resolution, with Nesterenko’s modular-function results [Nes96] the nearest proven algebraic-independence input; and Winkelmann’s Proposition 3 [Wi04, §3.2] for the same reduction seventeen years earlier. The unconditional refutation of (T2) has a second and stronger route as well: Barré-Sirieix, Diaz, Gramain and Philibert’s proof of the Mahler–Manin conjecture [BDGP96] gives $j(E_p)$ transcendental — the nome $q = p^{-1}$ is algebraic with $|q| < 1$ — hence no complex multiplication, hence $\text{End}(E_p) = \mathbf{Z}$. The published state of the art on (T1) is Roy [Roy92] and Diaz [Dia07], and the citable reason the proven machinery misses it is that Diaz’s Corollaire 1(1) needs $(\lambda_0, \lambda_2, \bar{\lambda}_2)$ to be $\overline{\mathbf{Q}}$ -linearly free, which fails here because $\lambda_2 = \log q$ is real.

Withdrawn. An earlier draft of this section read: “The question (T1) appears to be unrecorded in the literature in this form and is stated here as open.” **That is false and is withdrawn.** $\neg(\text{T1})$ is the specialization to $\alpha_1 = p$, $\alpha_2 = q$ of the conjecture printed as (C4E faible) in [Dia97, p. 231], due to D. Bertrand [Ber97, §5], and the same paper carries unconditional partial results this paper does not consume. The germ is older still: Alaoglu and Erdős asked the two-prime question in print in 1944 [AE44, pp. 449, 455]. The mathematics is untouched: (T1) is open, the family-level kill does not consume it, and (T2) is refuted unconditionally.

Novelty statement. The construction — the $E_p \times E_q$ assembly proposal, put forward and refuted here — is new. **Theorems 1, 2 and 3 are not: they are Winkelmann’s, from 15 April 2002, specialized here to primes** (item 8 above, and the attribution note at the head of §3). What is claimed as new is the prime-indexed explication of the rigidity — the group $\text{Hom}(E_p, E_q)$, its rank, and the isogeny degree uv , none of which appears in [Wi04] — together with Theorem 4 (Néron–Severi collapse and its four consequences), Theorem 5 (the prime-blind diagonal residue), Theorem 6 (the no-go), and the identification of the seed proposal as empty. The eight citation-and-distinguish obligations above are discharged. The obstruction’s ingredients are elementary — Birkenhake–Lange-level lattice arithmetic plus unique factorization — which is exactly why the rigidity half was available to be noticed in 2002, and was.

Provenance

Remark (Provenance). This paper is the publishable record of a result obtained on August 26, 2026 inside a structured research program on the Riemann hypothesis; the seed proposal refuted here (§1.3) originated inside the same program, where it is retained verbatim as an immutable record. The refutation was constructed independently by two parties under a duplicate-blind protocol (identical mathematics, no shared text), then re-derived in full by a third who had written neither — whose record documents the $4\pi^2\mathbf{Q}$ versus $\pi^2\mathbf{Q}$ reconciliation of Remark 1 — and

re-proved from scratch for this paper. The numerical checks cited in the text (check A: the 55-digit confirmation of Lemma 0; check B: the synthetic isogeny pair with $yy' = 3/7$; check C: the PSLQ/continued-fraction table of §7; check D: the kernel tabulations for $k \leq 7$; check E: the positive control $\log 8 = 3 \log 2$) are implemented in `results/c3-r/seed-no-go-checks.py` and `results/c3-r/seed-no-go-checks.json` at 400-digit working precision; a referee suite reproduced them bit-for-bit and re-implemented them independently with rigorous interval-arithmetic continued fractions and instrument-independent constants, with a third instrument as cross-check. An independent referee pass dated August 26, 2026 found one fatal defect, cutting in this paper’s own disfavor: the original version of §7 claimed that no proven theorem excludes either coincidence (T1)/(T2), whereas (T2) has been excluded unconditionally since 1934 by Gelfond–Schneider, and (T1) is settled already by the four-exponentials conjecture, far short of Schanuel. The repair — the Proposition of §7, the unconditional form of Theorem 5(1), and the corrected status ladder — is incorporated here; the consequence is that $\text{End}(E_p) = \mathbf{Z}$ for every prime with no hedge, leaving exactly one open rider, (T1). Two limitations of the literature search are recorded rather than glossed: Akatsuka’s paper [Ak09] is characterized through Tanaka’s introduction [Ta20] and venue confirmations, its full text being paywalled, and nothing load-bearing rests on unread portions; quotations from [CC26], [CC25] and [CE15] were checked against full texts.

Data and code availability. The numerical checks above, the referee record, the literature search and the correction notes referred to in this section are in the accompanying repository, <https://github.com/x67ai/riemann-rh-program>, at the tag `paper-2026-08-28`, which is the state this paper was prepared against; paths are relative to `rh-program/`. The working PDF corpus of fetched literature is deliberately not redistributed, for copyright reasons; nothing here depends on it, and every external claim is cited to its published source.

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